SO-ARM101 开源 6 轴机械臂使用文档

1. 虚拟机 linux 环境配置

虚拟机下载安装步骤（步骤 1-3.6）：

https://blog.csdn.net/m0_37383484/article/details/138029827

ubuntu22.04 系统镜像文件下载链接：https://releases.ubuntu.com/jammy/ubuntu-22.04.5-desktop-amd64.iso

2. 机械臂组装

请在输入舵机 ID 后观看组装视频组装，提前组装会导致 ID 输入时，接线困难

3. miniconda3 环境

3.1miniconda3 安装

> 按 CTRL+ALT+T 打开终端，输入更新软件包指令
>
> sudo apt update && sudo apt upgrade
>
> 
>
>
> 遇到该界面按 enter 表示继续
>
> 终端输入下载安装包：

wget https://mirrors.tuna.tsinghua.edu.cn/anaconda/miniconda/Miniconda3-py310_23.11.0-2-Linux-x86_64.sh

> 运行安装：bash ‘/home/主机名/Miniconda3-py310_23.11.0-2-Linux-x86_64.sh’
>
> 按下 enter 键表示确定（主机名是指自己的虚拟机主机名，例如此处作者使用的是 host）

```
host@111:~$ bash '/home/host/Miniconda3-py310_23.11.0-2-Linux-x86_64.sh'

Welcome to Miniconda3 py310_23.11.0-2

In order to continue the installation process, please review the license
agreement.
Please, press ENTER to continue
>>>
```

```
======================================
End User License Agreement - Miniconda
======================================

Copyright 2015-2023, Anaconda, Inc.

All rights reserved under the 3-clause BSD License:

This End User License Agreement (the "Agreement") is a legal agreement between y
ou and Anaconda, Inc. ("Anaconda") and governs your use of Miniconda.

Subject to the terms of this Agreement, Anaconda hereby grants you a non-exclusi
ve, non-transferable license to:

  * Install and use the Miniconda,
  * Modify and create derivative works of sample source code delivered in Minico
nda subject to the Terms of Service for the Repository (as defined hereinafter)
available at https://www.anaconda.com/terms-of-service, and
  * Redistribute code files in source (if provided to you by Anaconda as source)
 and binary forms, with or without modification subject to the requirements set
forth below.

Anaconda may, at its option, make available patches, workarounds or other update
--More--
```

按↓键，直到出现该界面后输入 yes，再按 enter

```
The following packages listed on https://www.anaconda.com/cryptography are inclu
ded in the Repository accessible through Miniconda that relate to cryptography.

Last updated March 21, 2022


Do you accept the license terms? [yes|no]
>>>
```

按 enter 确认

```
Do you accept the license terms? [yes|no]
>>> yes

Miniconda3 will now be installed into this location:
/home/host/miniconda3

  - Press ENTER to confirm the location
  - Press CTRL-C to abort the installation
  - Or specify a different location below

[/home/host/miniconda3] >>>
```

输入 yes，再按 enter

```
conda config --set auto_activate_base false

You can undo this by running `conda init --reverse $SHELL`? [yes|no]
[no] >>>
```

输入 conda --version 出现版本号即为成功

```
host@111:~$ conda --version
conda 23.11.0
```

3.2 miniconda 换源

关闭终端重新按 ctrl+alt+T 打开新终端，输入 conda config --set show_channel_urls yes 后输入 gedit .condarc

```
(base) host@111:~$ conda config --set show_channel_urls yes
(base) host@111:~$ gedit .condarc
```

输入以下源通道后按 ctrl+s 保存后退出

```
channels:
- defaults
show_channel_urls: true
default_channels:
- https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
- https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/r
- https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/msys2
custom_channels:
conda-forge: https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud
pytorch: https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud
```

```
Open    *.condarc    Save
        ~/
channels:
- defaults
show_channel_urls: true
default_channels:
- https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/main
- https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/r
- https://mirrors.tuna.tsinghua.edu.cn/anaconda/pkgs/msys2
custom_channels:
conda-forge: https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud
pytorch: https://mirrors.tuna.tsinghua.edu.cn/anaconda/cloud
```

终端输入 conda clean -i 清除缓存

```
(base) host@111:~$ gedit .condarc
(base) host@111:~$ conda clean -i
```

输入 y，再按 enter

```
Will remove 1 index cache(s).
Proceed ([y]/n)? y
```

3.3 虚拟环境配置

终端输入，创建虚拟环境：

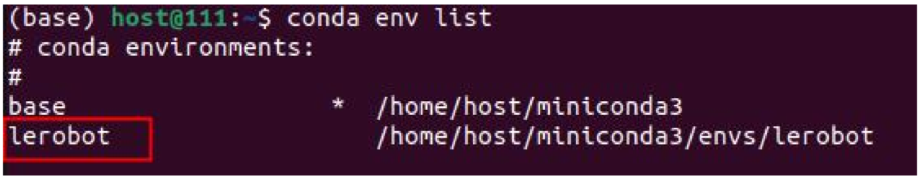
```
conda create -n lerobot python=3.10.18 ffmpeg=7.1.1 -c conda-forge
```

```
(base) host@111:~$ conda create -n lerobot python=3.10.18 ffmpeg=7.1.1 -c conda-
forge
Channels:
 - conda-forge
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done
```

输入 y，再按 enter

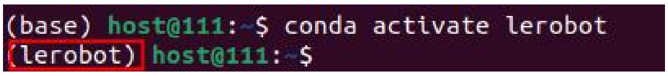
```
  xorg-libxfixes     conda-forge/linux-64::xorg-libxfixes-6.0.2-hb03c661_0
  xorg-libxrandr     conda-forge/linux-64::xorg-libxrandr-1.5.4-hb9d3cd8_0
  xorg-libxrender    conda-forge/linux-64::xorg-libxrender-0.9.12-hb9d3cd8_0
  xorg-libxscrnsaver conda-forge/linux-64::xorg-libxscrnsaver-1.2.4-hb9d3cd8_
  zstd               conda-forge/linux-64::zstd-1.5.7-h3691f8a_5


Proceed ([y]/n)?
```

终端输入 conda env list，出现 lerobot 则创建成功



```
(base) host@111:~$ conda env list
# conda environments:
#
base                  *  /home/host/miniconda3
lerobot                  /home/host/miniconda3/envs/lerobot
```

终端输入 conda activate lerobot 进入环境，主机名前出现 lerobot 或 base 改为 lerobot 则进入成功

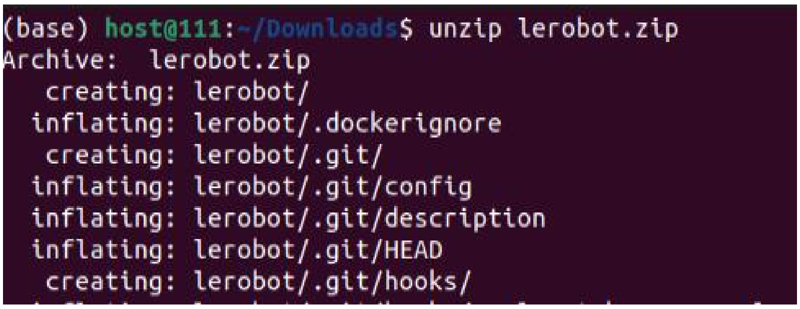


```
(base) host@111:~$ conda activate lerobot
(lerobot) host@111:~$
```

下载并移动 lerobot 程序源码(压缩包中有)：

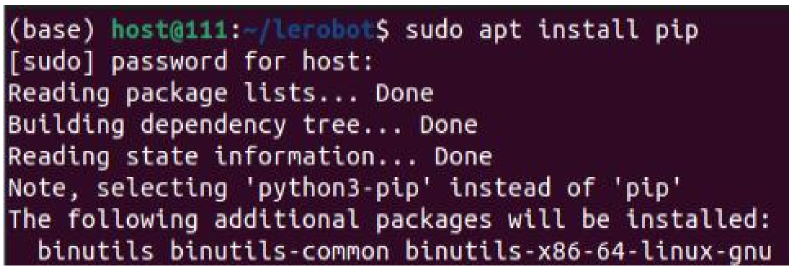
将源码文件拖入或复制粘贴入虚拟机(放置在共享文件夹也可以)，不能复制请重新安装 open-vm-tools-desktop

然后使用 **unzip lerobot.zip** 解压



```
(base) host@111:~/Downloads$ unzip lerobot.zip
Archive:  lerobot.zip
   creating: lerobot/
  inflating: lerobot/.dockerignore
   creating: lerobot/.git/
  inflating: lerobot/.git/config
  inflating: lerobot/.git/description
  inflating: lerobot/.git/HEAD
   creating: lerobot/.git/hooks/
```

输入 **cd lerobot** 进入文件夹，再输入 **sudo apt install pip** 安装 pip



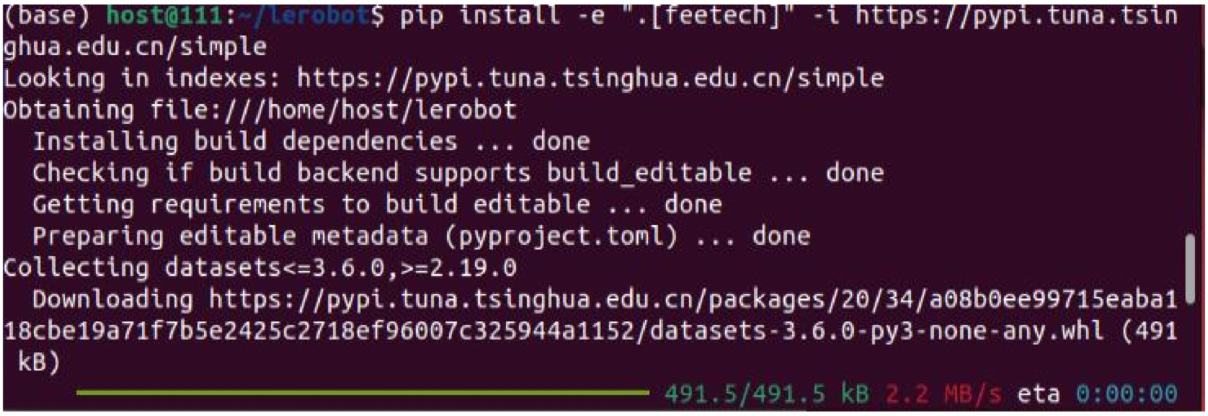
```
(base) host@111:~/lerobot$ sudo apt install pip
[sudo] password for host:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'python3-pip' instead of 'pip'
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu
```

输入：conda install ffmpeg=7.1.1 -c conda-forge

```
(lerobot) host@111:~/lerobot$ conda install ffmpeg=7.1.1 -c conda-forge
Channels:
 - conda-forge
 - defaults
Platform: linux-64
Collecting package metadata (repodata.json): done
Solving environment: done
```

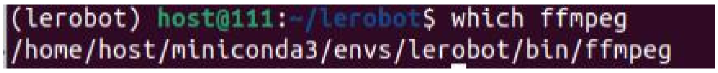
终端输入安装依赖包和驱动：

pip install -e ".[feetech]" -i https://pypi.tuna.tsinghua.edu.cn/simple



```
(base) host@111:~/lerobot$ pip install -e ".[feetech]" -i https://pypi.tuna.tsin
ghua.edu.cn/simple
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Obtaining file:///home/host/lerobot
  Installing build dependencies ... done
  Checking if build backend supports build_editable ... done
  Getting requirements to build editable ... done
  Preparing editable metadata (pyproject.toml) ... done
Collecting datasets<=3.6.0,>=2.19.0
  Downloading https://pypi.tuna.tsinghua.edu.cn/packages/20/34/a08b0ee99715eaba1
18cbe19a71f7b5e2425c2718ef96007c325944a1152/datasets-3.6.0-py3-none-any.whl (491
 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 491.5/491.5 kB 2.2 MB/s eta 0:00:00
```

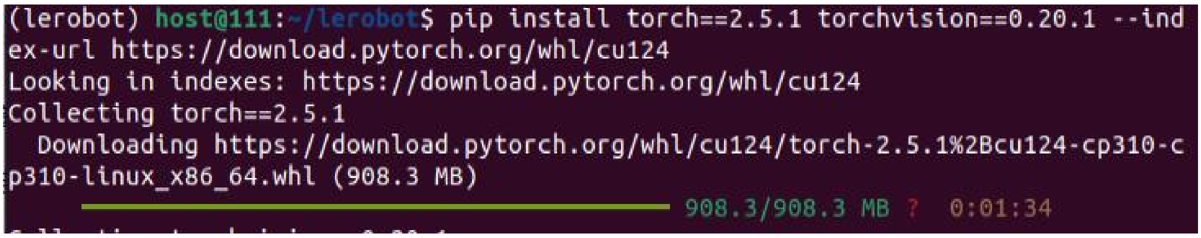
出现路径则安装成功



```
(lerobot) host@111:~/lerobot$ which ffmpeg
/home/host/miniconda3/envs/lerobot/bin/ffmpeg
```

安装 torch 和 torchvision：

pip install torch==2.5.1 torchvision==0.21.10 --index-url
https://download.pytorch.org/whl/cu124



```
(lerobot) host@111:~/lerobot$ pip install torch==2.5.1 torchvision==0.20.1 --ind
ex-url https://download.pytorch.org/whl/cu124
Looking in indexes: https://download.pytorch.org/whl/cu124
Collecting torch==2.5.1
  Downloading https://download.pytorch.org/whl/cu124/torch-2.5.1%2Bcu124-cp310-c
p310-linux_x86_64.whl (908.3 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 908.3/908.3 MB ? 0:01:34
```

4.舵机 id 发送

将 usb 线插入驱动板(从臂)并连接上 5V 电源适配器

插入 usb 后输入(如果没有检测到 usb，请将 usb 线转动 180° 再插入)：

python -m lerobot.find_port

```
(lerobot) host@111:~/lerobot$ python -m lerobot.find_port
Finding all available ports for the MotorsBus.
Ports before disconnecting: ['/dev/ttyACM0', '/dev/ttyprintk', '/dev/ttyS31', '/dev/ttyS
30', '/dev/ttyS29', '/dev/ttyS28', '/dev/ttyS27', '/dev/ttyS26', '/dev/ttyS25', '/dev/tt
yS24', '/dev/ttyS23', '/dev/ttyS22', '/dev/ttyS21', '/dev/ttyS20', '/dev/ttyS19', '/dev/
ttyS18', '/dev/ttyS17', '/dev/ttyS16', '/dev/ttyS15', '/dev/ttyS14', '/dev/ttyS13', '/de
v/ttyS12', '/dev/ttyS11', '/dev/ttyS10', '/dev/ttyS9', '/dev/ttyS8', '/dev/ttyS7', '/dev
/ttyS6', '/dev/ttyS5', '/dev/ttyS4', '/dev/ttyS3', '/dev/ttyS2', '/dev/ttyS1', '/dev/tty
S0', '/dev/tty63', '/dev/tty62', '/dev/tty61', '/dev/tty60', '/dev/tty59', '/dev/tty58',
 '/dev/tty57', '/dev/tty56', '/dev/tty55', '/dev/tty54', '/dev/tty53', '/dev/tty52', '/d
ev/tty51', '/dev/tty50', '/dev/tty49', '/dev/tty48', '/dev/tty47', '/dev/tty46', '/dev/t
ty45', '/dev/tty44', '/dev/tty43', '/dev/tty42', '/dev/tty41', '/dev/tty40', '/dev/tty39
', '/dev/tty38', '/dev/tty37', '/dev/tty36', '/dev/tty35', '/dev/tty34', '/dev/tty33', '
/dev/tty32', '/dev/tty31', '/dev/tty30', '/dev/tty29', '/dev/tty28', '/dev/tty27', '/dev
/tty26', '/dev/tty25', '/dev/tty24', '/dev/tty23', '/dev/tty22', '/dev/tty21', '/dev/tty
20', '/dev/tty19', '/dev/tty18', '/dev/tty17', '/dev/tty16', '/dev/tty15', '/dev/tty14',
 '/dev/tty13', '/dev/tty12', '/dev/tty11', '/dev/tty10', '/dev/tty9', '/dev/tty8', '/dev
/tty7', '/dev/tty6', '/dev/tty5', '/dev/tty4', '/dev/tty3', '/dev/tty2', '/dev/tty1', '/
dev/tty0', '/dev/tty']
Remove the USB cable from your MotorsBus and press Enter when done.
```

拔出 usb 后按 enter，/dev/ttyACM0 为从臂 usb 驱动名(usb 默认为先后顺序排列驱动名，一般先插为/dev/ttyACM0,第二个则为/dev/ttyACM1，以此类推，请注意，驱动名也有可能是/dev/ttyUSB0，如果是这样，请把命令中 ACM 部分改为 USB)

```
Remove the USB cable from your MotorsBus and press Enter when done.

The port of this MotorsBus is '/dev/ttyACM0'
Reconnect the USB cable.
```

终端输入给予驱动端口权限：sudo chmod 666 /dev/ttyACM0

输入主臂 id 则输入(红色部分为需要根据驱动名更改的部分)：

```
lerobot-setup-motors \
    --teleop.type=so101_leader \
    --teleop.port=/dev/ttyACM0
```

输入从臂 id 则输入：

```
lerobot-setup-motors \
    --teleop.type=so101_follower \
    --teleop.port=/dev/ttyACM0
```

按从高到低分别连接六个舵机(夹爪->底盘旋转)，每次确认连接好舵机线和 usb 线以及数据线后按 enter 即可赋 id，**赋 id 时舵机与舵机之间禁止连接，一次只可输入一个舵机 ID 号**

```
(lerobot) host@111:~/lerobot$ lerobot-setup-motors     --robot.type=so101_follow
er     --robot.port=/dev/ttyACM0
Connect the controller board to the 'gripper' motor only and press enter.
'gripper' motor id set to 6
Connect the controller board to the 'wrist_roll' motor only and press enter.
'wrist_roll' motor id set to 5
Connect the controller board to the 'wrist_flex' motor only and press enter.
'wrist_flex' motor id set to 4
Connect the controller board to the 'elbow_flex' motor only and press enter.
'elbow_flex' motor id set to 3
Connect the controller board to the 'shoulder_lift' motor only and press enter.
'shoulder_lift' motor id set to 2
Connect the controller board to the 'shoulder_pan' motor only and press enter.
'shoulder_pan' motor id set to 1
```

Tips：从臂赋 id 号过程同上

5. 校准机械臂

```
sudo chmod 666 /dev/ttyACM*
rm -rf ~/.cache/huggingface/lerobot/calibration/robots
rm -rf ~/.cache/huggingface/lerobot/calibration/teleoperators

lerobot-calibrate \
    --robot.type=so101_follower \
    --robot.port=/dev/ttyACM0 \
    --robot.id=my_awesome_follower_arm
lerobot-calibrate \
    --teleop.type=so101_leader \
    --teleop.port=/dev/ttyACM1 \
    --teleop.id=my_awesome_leader_arm
```

将机械臂各关节调至中间位置，按下 enter，然后活动每个关节直至最大最小范围后按 enter 保存(如果发现 usb 连接不上，请把 usb 反过来插)

```
(lerobot) seeed@seeed-KuangshiG16-PRO-Series-GM6PQ7Q:~/lerobot$ python -m lerobo
t.calibrate     --robot.type=so101_follower     --robot.port=/dev/ttyACM0 --robo
t.id=my_awesome_follower_arm
INFO 2025-08-09 14:54:41 calibrate.py:73 {'robot': {'calibration_dir': None,
           'cameras': {},
           'disable_torque_on_disconnect': True,
           'id': 'my_awesome_follower_arm',
           'max_relative_target': None,
           'port': '/dev/ttyACM0',
           'use_degrees': False},
 'teleop': None}
INFO 2025-08-09 14:54:41 follower.py:104 my_awesome_follower_arm SO101Follower c
onnected.
INFO 2025-08-09 14:54:41 follower.py:121
Running calibration of my_awesome_follower_arm SO101Follower
Move my_awesome_follower_arm SO101Follower to the middle of its range of motion
and press ENTER....
```

6. 遥感控制

该指令为从臂 usb 驱动名为/dev/ttyACM0,主臂 usb 驱动名为/dev/ttyACM1，如果不同请更改指令 port 部分

```
sudo chmod 666 /dev/ttyACM*
lerobot-teleoperate \
    --robot.type=so101_follower \
    --robot.port=/dev/ttyACM0 \
    --robot.id=my_awesome_follower_arm \
    --teleop.type=so101_leader \
    --teleop.port=/dev/ttyACM1 \
    --teleop.id=my_awesome_leader_arm
```

```
(lerobot) host@111:~/lerobot$ lerobot-teleoperate     --robot.type=so101_follower     --robot.port=/dev/ttyACM0     --robot.id=my_awesome_follower_arm     --teleop.type=so101_leader     --teleop.port=/dev/ttyACM1     --teleop.id=my_awesome_leader_arm
INFO 2025-12-04 11:13:56 eoperate.py:135 {'display_data': False,
 'fps': 60,
 'robot': {'calibration_dir': None,
           'cameras': {},
           'disable_torque_on_disconnect': True,
           'id': 'my_awesome_follower_arm',
           'max_relative_target': None,
           'port': '/dev/ttyACM0',
           'use_degrees': False},
 'teleop': {'calibration_dir': None,
            'id': 'my_awesome_leader_arm',
            'port': '/dev/ttyACM1',
            'use_degrees': False},
 'teleop_time_s': None}
INFO 2025-12-04 11:13:56 01_leader.py:82 my_awesome_leader_arm SO101Leader connected.
INFO 2025-12-04 11:13:56 follower.py:104 my_awesome_follower_arm SO101Follower connected.
```

做到这一步，就可以开始通过控制主臂来控制从臂了，如要闭关进程，关闭 终端或 Ctrl+C 即可